

Quantum Track Reconstruction in the LHCb VELO

From the Hamiltonian to HHL to the 1-Bit Quantum Filter

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Based on TrackHHL: [arXiv:2511.11458](https://arxiv.org/abs/2511.11458) (HHL) · [arXiv:2601.07766](https://arxiv.org/abs/2601.07766) (1-Bit Quantum Filter)

Outline

- 1 Motivation
- 2 The toy model
- 3 Hamiltonian & matrix inversion
- 4 Quantum inversion: $\text{HHL} \rightarrow \text{1-bit} \rightarrow \text{1BQF}$
- 5 Results from the toy
- 6 Next steps: QSVT

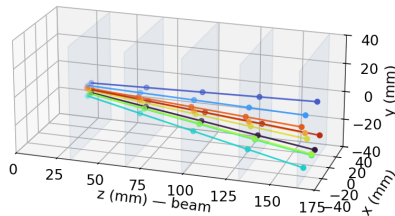
The problem: tracking is a combinatorial needle-in-a-haystack

- The LHCb **VELO** reconstructs charged particles (**tracks**) from **hits** on silicon planes.
- HL-LHC pile-up $\Rightarrow$ reconstruction cost threatens to outpace classical resources.
- Work with **segments** (hit $\rightarrow$ hit links on adjacent planes). With T hits/plane each gap offers T^2 candidates, only T real:

$$n_{\text{seg}} = 4T^2 \quad \text{vs} \quad n_{\text{true}} = 4T \quad (\text{signal fraction } 1/T).$$

- At $T=400$: **640,000** segments, **1,600** true (0.25%).
- **Idea:** cast track finding as a *linear system* $Ax = b$ and solve it classically *or* on a quantum computer.

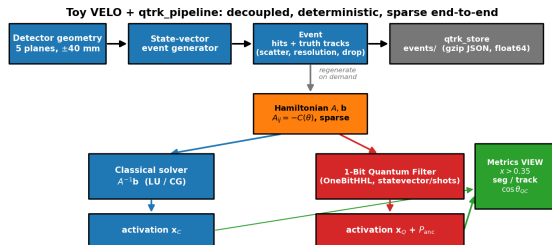
Toy VELO event: 10 tracks through 5 planes ($\phi_{\text{max}} = 0.2$)



A generated toy VELO event: tracks fan from the PV through 5 planes.

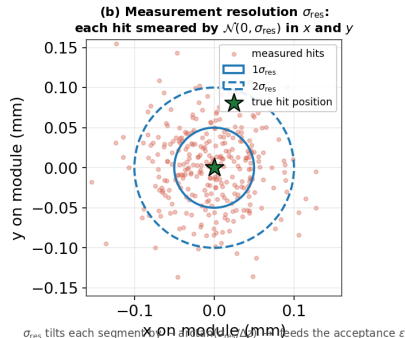
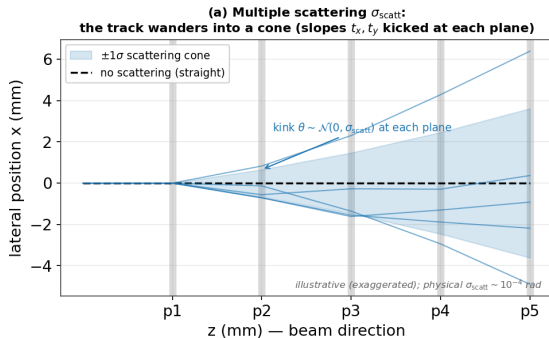
A controlled toy VELO — our testbed

- LHCb_VeLo_Toy_Model: **5 planes** at $z=33, 66, 99, 132, 165$ mm, ± 40 mm.
- Deterministic, fully controlled: sweep multiplicity T , noise, density.
- Decoupled qtrk_pipeline: events stored once; matrix A regenerated on demand (sparse); metrics recomputed as a *view*.
- Lets us isolate *how each factor* (noise, density, T , kernel) moves the algorithm.



The toy injects the two physical detector noises

The two detector noises the toy injects — both widen the triplet kink angle, set by $\varepsilon = \sqrt{2(s\sigma_{\text{scatt}})^2 + 12\arctan^2(s\sigma_{\text{res}}/\Delta z) + \dots}$



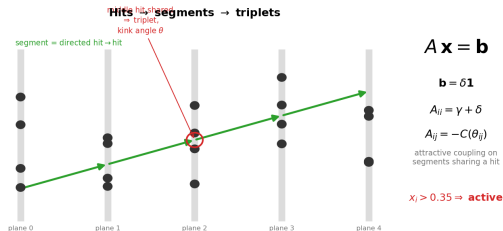
- **Multiple scattering σ_{scatt} :** kicks the slopes (t_x, t_y) at each plane $\Rightarrow$ the track wanders into a cone.
- **Resolution σ_{res} :** smears each hit in x, y . Both *widen the triplet kink angle*.

The Hamiltonian: an Ising-like segment model

- Two segments sharing a middle hit form a **triplet** with kink angle θ ; small $\theta \Rightarrow$ same particle.
- Assign each segment an activation x_i and minimise an Ising-like (Hopfield/Denby–Peterson) energy

$$H(\mathbf{x}) = -\frac{1}{2} \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x}.$$

- Off-diagonal coupling *rewards* compatible triplets; diagonal penalises activation.



From the Hamiltonian to a matrix inversion

Stationarity $\partial H/\partial \mathbf{x} = 0$ gives a **linear system**:

$$\boxed{A\mathbf{x} = \mathbf{b} = \delta\mathbf{1}} \quad \mathbf{x} = A^{-1}\mathbf{b}.$$

$$A_{ii} = \gamma + \delta, \quad A_{ij} = -C(\theta_{ij}),$$

with C the 0/1 compatibility adjacency ($\theta < \varepsilon$), so

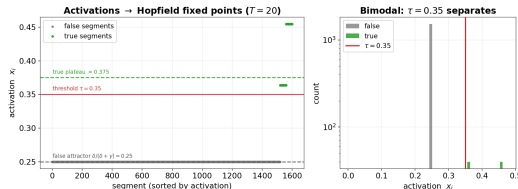
$$A = (\gamma + \delta)I - C, \quad \gamma=3, \delta=1.$$

Hopfield fixed points $\Rightarrow$ an absolute operating point:

- false/isolated $\rightarrow \delta/(\delta + \gamma) = 0.25$;
- true (in a chain) $\rightarrow \approx 0.375$ plateau;
- threshold $\tau = 0.35$ separates them.

Classical solve = state-of-the-art efficiency

[arXiv:2511.11458].

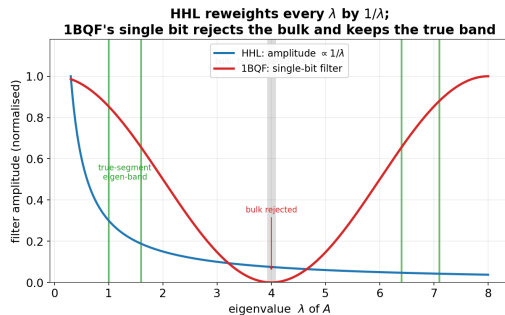


HHL: invert A on a quantum computer

Harrow–Hassidim–Lloyd solves $Ax = b$:

- 1 prepare $|b\rangle$;
- 2 **quantum phase estimation** writes each eigenvalue λ onto n_{time} time qubits;
- 3 controlled rotation scales each eigencomponent by $1/\lambda$;
- 4 post-select an ancilla; uncompute.

Promises an **exponential speedup in the number of hits** — *contingent* on efficient QPE and reading out the result. **Catch:** dense e^{iAt} and high-precision QPE $\Rightarrow$ prohibitive circuit depth on NISQ hardware.

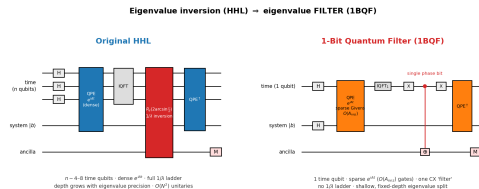


HHL reweights every eigenvalue by $1/\lambda$.

TrackHHL: restrict phase estimation to one bit

[arXiv:2511.11458 — Chiotopoulos, Lucio Martinez, Nicotra *et al.*]

- For this Hamiltonian the spectrum is **bimodal** — a single bit of eigenvalue precision is enough.
- Restricting QPE to **one bit**:
 - cuts circuit depth by up to 10^4 ;
 - resolves HHL's read-out problem;
 - enables a primary-vertex post-processing step.
- Classical reconstruction efficiency is retained.



The 1-Bit Quantum Filter (1BQF)

[arXiv:2601.07766 — Chiotopoulos, Nicotra, Scriven *et al.*]

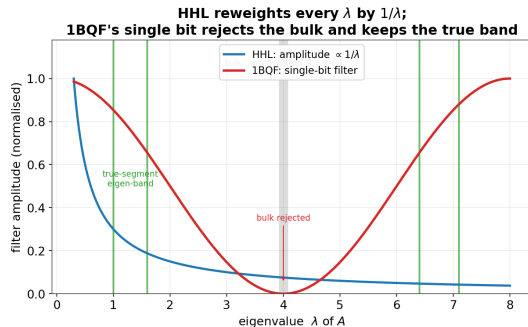
Reframe: matrix inversion $\rightarrow$ binary ground-state filtering.

- One time qubit + a single ancilla flip realises the filter

$$w_Q(\lambda) = \cos\left(\frac{\pi}{2} \frac{\lambda}{\gamma + \delta}\right),$$

whose *only* zero is the **notch** at $\lambda = \gamma + \delta$.

- Sparse e^{iAt} (two-level/Givens) $\Rightarrow$ gate complexity $\mathcal{O}(\sqrt{N} \log N)$.
- Runs on Quantinuum H2 and IBM Heron noise models.



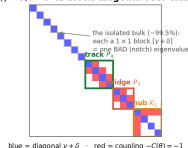
1BQF keeps a band with a single notch.

Why one bit suffices: the spectrum is block-diagonal

- $A = (\gamma + \delta)I - C$ is **block-diagonal over clusters**; eigenvalues $\lambda = (\gamma + \delta) - \mu(C)$.
- **bad (notch) eigenvalue**
 $\iff \mu(C) = 0 \iff \lambda = \gamma + \delta$.
- 99.5% of segments are **isolated false** (1×1 block at the notch) $\Rightarrow$ the filter erases them for free.
- True 4-chains sit *off* the notch $\Rightarrow$ kept.

Encoding the errors in A : one block per cluster type — good (off-notch) vs bad (notch) eigenvalues

$A = (\gamma + \delta)I - C$ is block-diagonal over clusters



From the Hamiltonian to the spectrum

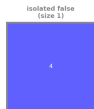
$$A\mathbf{x} = \mathbf{b} = \delta\mathbf{1}, \quad A_\theta = \gamma + \delta, \quad A_\theta = -C(\theta_0)$$

$C = 0/1$ compatibility adjacency (share a hit, $\theta < \epsilon$)

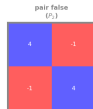
$$\Rightarrow A = (\gamma + \delta)I - C \quad (\text{block-diagonal over clusters})$$

each cluster's eigenvalues: $\lambda = (\gamma + \delta) - \mu(C)$

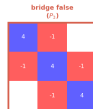
- $\mu(C) = 0 \iff \lambda = \gamma + \delta$ = the NOTCH = BAD
- isolated segment = $[\gamma + \delta] \rightarrow$ sits on the notch
- a hub $K_{1,m}$ has $m - 1$ bad eigenvalues
- quantum 1-bit filter $f(\lambda) = \cos(\frac{\pi}{2\gamma + \delta} \lambda)$ erases the notch ($f = 0$), keeps everything off it



$\lambda = \{4.00\}$
1 bad (notch)
quantum: erased



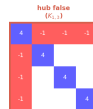
$\lambda = \{3.00, 5.00\}$
0 bad (notch)
quantum: rejected



$\lambda = \{2.59, 4.00, 5.41\}$
1 bad (notch)
quantum: survives



$\lambda = \{2.38, 3.38, 4.62, 5.62\}$
0 bad (notch)
quantum: kept

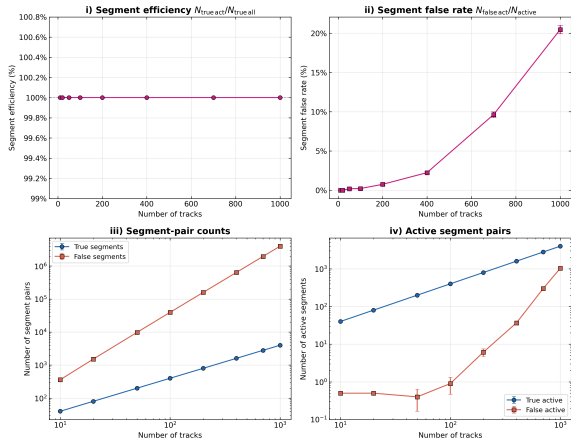


$\lambda = \{2.27, 4.00, 4.00, 5.73\}$
2 bad (notch)
quantum: survives

Segment efficiency stays high — the false rate is the story

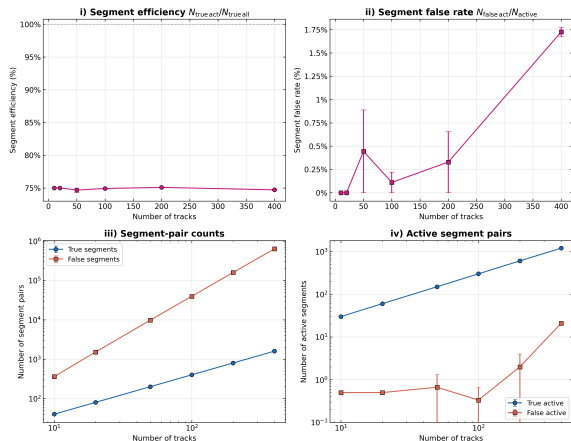
Classical

Classical solver — segment metrics ($\epsilon=2$ mrad, $\gamma=3$, no hit drop)



1BQF (quantum)

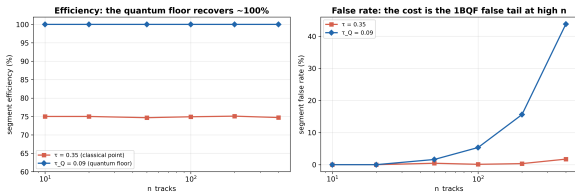
Quantum (1BQF) solver — segment metrics ($\epsilon=2$ mrad, $\gamma=3$, no hit drop)



Efficiency $\approx 100\%$ (classical) / $\approx 75\%$ (1BQF) across multiplicity; the **false rate grows with T** — the rare coupled cross-track clusters.

The efficiency–false-rate trade-off (the real result)

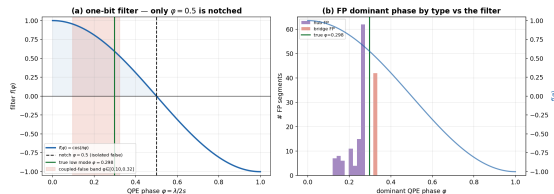
Operating-point trade-off — both achievable up to moderate T , then the false tail bites



- You *can* push segment efficiency high — but only by **lowering the threshold**, which admits false positives.
- At $\tau=0.35$: 1BQF eff = 75%; to recover $\approx 100\%$ the false rate climbs to $\approx 44\%$.
- **High efficiency and high false rate come together** — not low/low. The operating curve, not a single point, is the result.

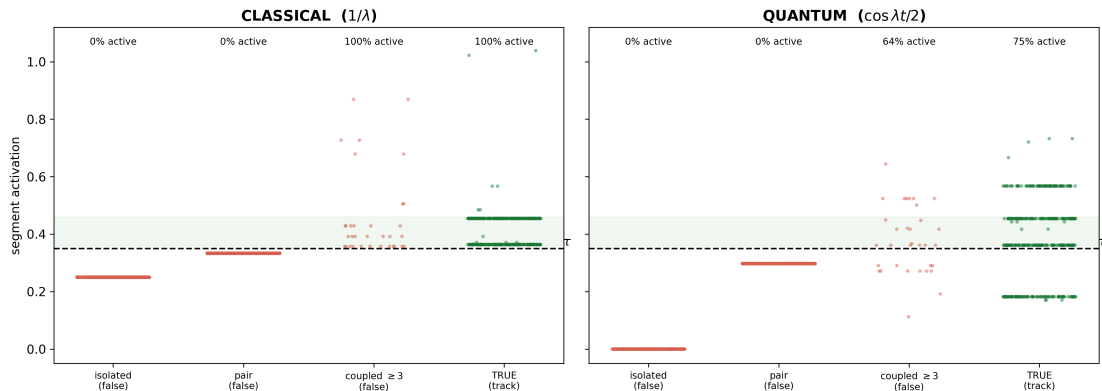
Why the trade-off is fundamental

- The single notch removes only the **isolated** false class.
- Surviving false positives are **coupled cross-track bridges/hubs** — they carry *good*, off-notch eigenvalues.
- In the QPE phase the hub/bridge modes fall in the **same band as true segments** (one zero can't separate them).
- **Topological degeneracy**: a false chain of the same shape as a true track has an identical A-block $\Rightarrow$ no threshold (classical or quantum) can separate them.



Class-by-class: where each segment lands vs τ

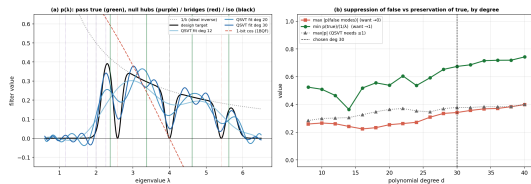
Treatment of every segment class: where it lands vs τ (green=true, red=false; band=true-track range)



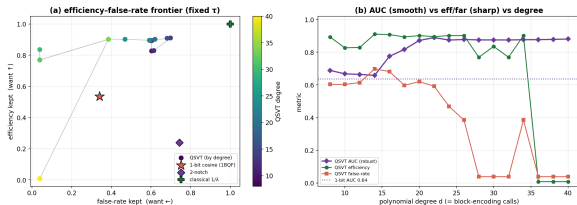
Isolated/pair false $\rightarrow$ below τ (both solvers); coupled- ≥ 3 false $\rightarrow$ above τ (false positives, both); true tracks $\rightarrow$ above τ (1BQF drops the outer segment $\rightarrow$ eff ≈ 0.75).

Beyond one notch: Quantum Singular Value Transformation

- The 1BQF is QSVT restricted to $p = \cos$ — **one** notch.
- **QSVT** (Gilyén–Su–Low–Wiebe) realises *any* bounded polynomial $p(A)$ from a block-encoding of $A + d$ phase rotations.
- Target: a **band-limited inverse** — $p(\lambda) \approx 1/\lambda$ on true-track eigenvalues, $p(\lambda) \approx 0$ on every false mode (*many* notches).
- Works *iff* false and true eigenvalues are separated — and on real $T=200$ events the false modes sit ≥ 0.2 from every true eigenvalue.



QSVT: a polynomial filter lifts true/false separation



- A degree- ~ 30 polynomial lifts true/false **AUC** from 0.49 (classical) / 0.64 (1-bit) to **0.88**.
- Nulls *all* off-length bridges and $\sim 95\%$ of hubs.
- **Cost:** ~ 30 block-encoding calls $\times$ a $\sim 2\times$ amplitude-amplification overhead $\approx$ tens $\times$ the 1-bit depth; sparse- A preserved.
- **Irreducible floor:** a false bridge of the *same length* as a true track (0% at $T \leq 200$; grows with density).

Summary

- Track finding $\rightarrow$ Ising-like Hamiltonian $\rightarrow$ **matrix inversion** $Ax = \delta \mathbf{1}$, $A = (\gamma + \delta)I - C$.
- **HHL** solves it but needs deep circuits; **TrackHHL** restricts QPE to one bit (10^4 depth cut); the **1BQF** reframes it as a single-notch spectral filter, $O(\sqrt{N} \log N)$, NISQ-ready.
- On the toy: **segment efficiency stays high**; the false rate is set by coupled cross-track clusters — a **fundamental trade-off** (topological degeneracy), not a solver artefact.
- **QSVT** generalises the one notch to a polynomial filter: AUC $\rightarrow 0.88$, down to a same-length-bridge floor.

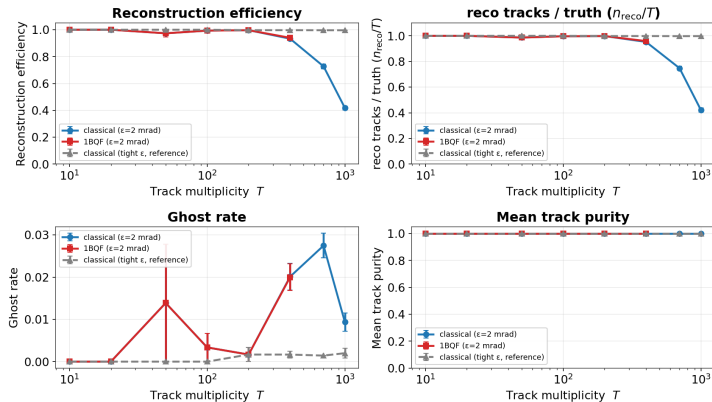
arXiv:2511.11458 (HHL) · arXiv:2601.07766 (1BQF) · toy: LHCb_VeLo_Toy_Model

Thank you

Questions?

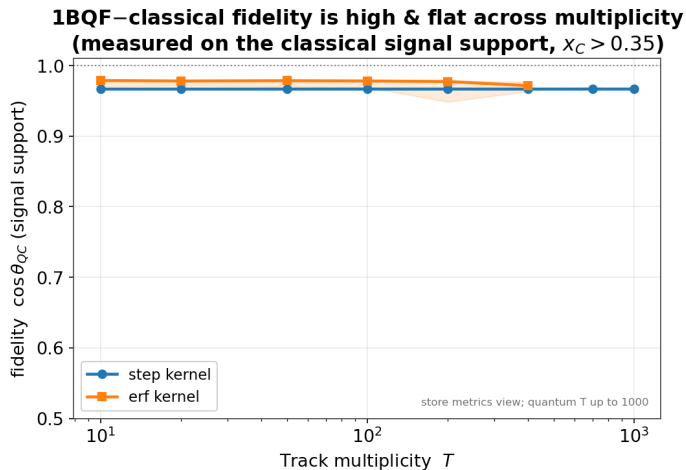
Backup: track-level metrics — segments propagate to tracks

Track-level LHCb metrics: segment false-positive bridges merge tracks at high T
(wide $\varepsilon=2$ mrad collapses efficiency; tight ε stays perfect — same events)



Wide $\varepsilon \Rightarrow$ bridges merge tracks $\Rightarrow$ reconstruction efficiency collapses ($1.0 \rightarrow 0.42$); tight ε stays perfect.

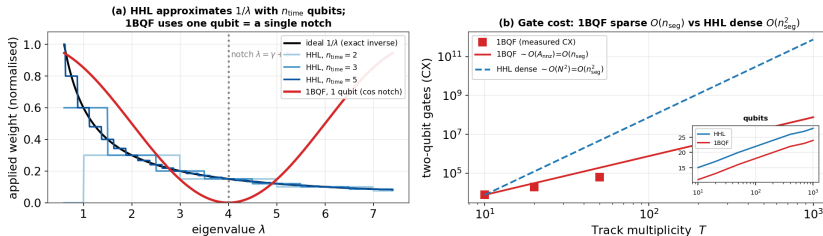
Backup: 1BQF—classical fidelity vs multiplicity



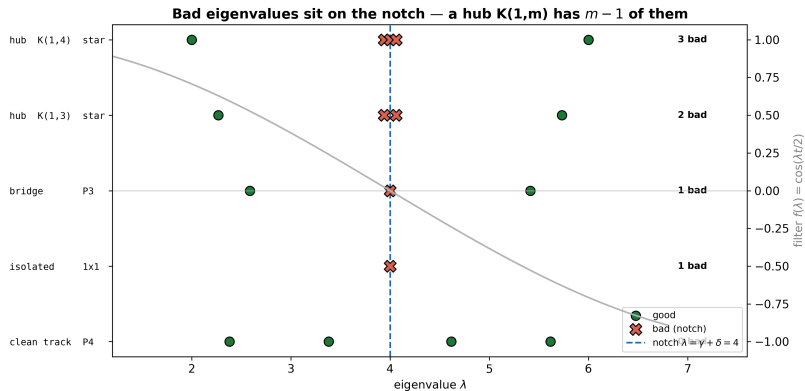
On the classical signal support the 1BQF is faithful: $\cos \theta_{QC} \approx 0.97$, flat to $T=1000$.

Backup: why one bit — cost vs full HHL

Why one bit: HHL needs an n_{time} -qubit eigenvalue register and a dense e^{iAt} ;
the 1BQF needs one qubit + a sparse e^{iAt} , and the single notch already separates the bimodal track spectrum

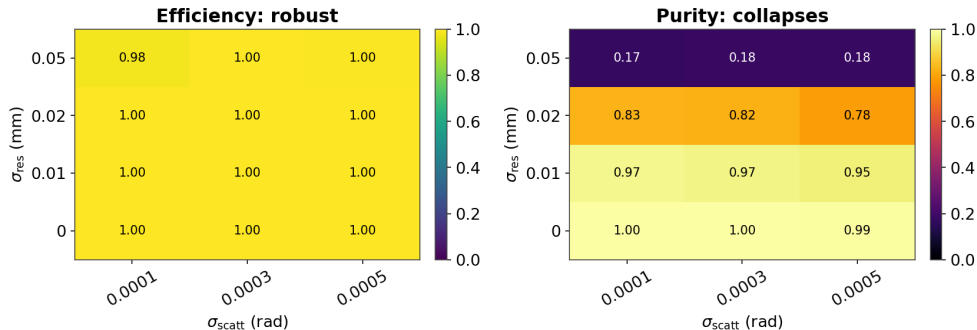


Backup: the bad-eigenvalue ladder



Backup: classical failure mode under noise

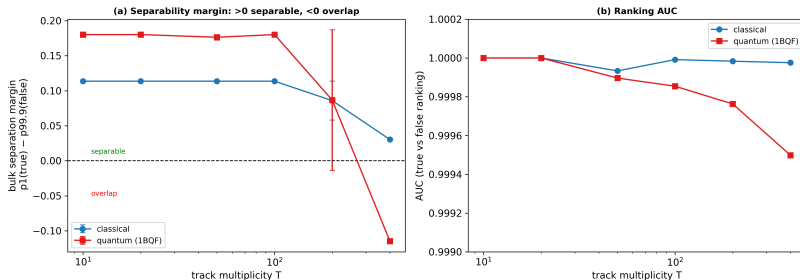
Classical failure mode at $T = 200$: resolution smearing spares efficiency but collapses purity



ε scales with noise $\Rightarrow$ efficiency robust, but **purity** collapses under resolution smearing.

Backup: separability vs multiplicity

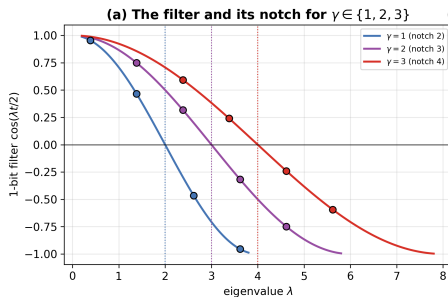
Separability vs track multiplicity (fixed- ϵ , $\gamma=3$, drop=0): quantum starts cleaner, degrades faster



Quantum starts cleaner, degrades faster; margin negative $\sim T=400$ — the coupled-false ceiling.

Backup: γ self-similarity of the notch

Self-similarity in γ : the chain spectrum is centred on $\gamma + \delta$, so the 1-bit notch always splits true segments into two plateaus



(b) Rescaled by the notch: one universal filter, modes always straddle it

